

Sequential Graph Neural Networks for Traffic Speed Prediction

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Abstract

Accurate short-term traffic speed prediction is a fundamental problem in intelligent transportation systems, directly impacting congestion management, route planning, and urban mobility optimization. Traditional time-series models and convolutional approaches often fail to capture the complex topological dependencies inherent in road networks. In this project, we implement and evaluate a Sequential Graph Neural Network (SeqGNN) framework that models traffic dynamics as a sequence of attributed graphs. Each road segment is represented as a node, while physical connectivity between segments is modeled through directed edges, enabling explicit learning of congestion propagation along network topology.

Our implementation combines Graph Network (GN) blocks with recurrent modeling through a Graph Recurrent Neural Network (GRNN), resulting in a graph-to-graph sequence learning architecture. The model is evaluated on the real-world METR-LA dataset and compared against strong baselines, including a Multilayer Perceptron (MLP), Nested LSTM (NLSTM), Seq2Seq LSTM, and Dual-Stage Attention RNN (DA-RNN). Performance is measured using MAE, RMSE, and MAPE.

1 Introduction

Traffic speed prediction is a central task in intelligent transportation systems, enabling proactive congestion mitigation, travel time estimation, and traffic signal optimization. Urban road networks exhibit complex spatiotemporal dependencies where traffic states evolve dynamically and propagate through network connectivity rather than simple spatial proximity (Figure 1 shows an snapshot of traffic in Los Angeles). Classical statistical models and conventional deep learning approaches, such as LSTMs or CNNs, struggle to represent these dependencies effectively.

Recent advances in graph mining and graph neural networks (GNNs) have provided powerful tools for modeling relational data. By representing road networks as graphs, GNNs can explicitly encode topological structure and learn how traffic conditions spread through connected road segments. However, most existing GNN-based traffic models either produce single-step predictions or fail to generate graph-structured outputs over time.

This project implements a Sequential Graph Neural Network (SeqGNN), which integrates sequence-to-sequence learning with graph neural networks. In this formulation, both the input and output are modeled as sequences of graphs, where node attributes correspond to traffic speeds and static road features, and edge attributes capture connectivity information. Temporal dependencies are modeled through Graph Recurrent Neural Network (GRNN) blocks, which extend classical RNNs to graph-structured data.

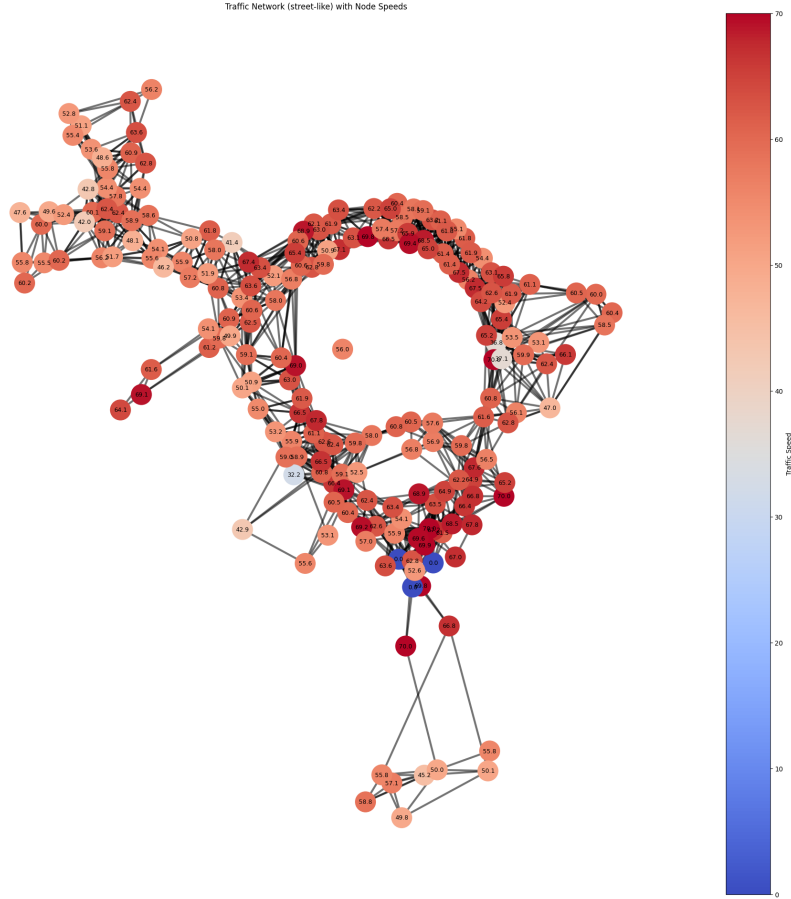


Figure 1: snapshot of traffic in Los Angeles

Compared to the original proposal, this implementation extends the experimental evaluation by incorporating multiple strong temporal baselines (MLP, NLSTM, Seq2Seq, and DA-RNN) under a unified training and evaluation pipeline. The objective of this project is to assess whether explicit graph modeling provides measurable benefits over purely temporal models on a real-world dataset.

2 Methodology

2.1 Dataset Description

Experiments are conducted on the METR-LA dataset, which contains traffic speed measurements collected from loop detectors in the Los Angeles highway network. The dataset consists of approximately 207 sensors (nodes) connected by directed edges derived from the road network topology. Traffic speeds are recorded at 5-minute intervals.

Table 1: METR-LA Dataset Statistics

Property	Value	Description
Number of Nodes	207	Traffic sensors
Number of Edges	~1,500	Road connections
Sampling Interval	5 minutes	Temporal resolution
Input Length	24	Past 2 hours
Output Length	12	Future 1 hour

Preprocessing includes missing value imputation via forward/backward filling, normalization, and sliding-window segmentation.

2.2 Algorithms and Models

2.2.1 Baseline Models

- **MLP**: A fully connected network treating each sensor independently.
- **NLSTM**: Nested LSTM with hierarchical memory for temporal modeling.
- **Seq2Seq**: Encoder-decoder LSTM for multi-step forecasting.
- **DA-RNN**: Dual-stage attention model capturing temporal importance.

2.2.2 SeqGNN Architecture

The proposed SeqGNN consists of:

- **GN Block**: Updates edge, node, and global features using message passing.
- **GRNN Block**: Extends GN blocks with recurrent hidden states.
- **Encoder**: Processes historical graph sequences.
- **Decoder**: Generates future graph sequences autoregressively.

Static node attributes (e.g., degree) and edge attributes (connectivity strength) are embedded and concatenated with dynamic traffic speeds.

2.3 Experimental Setup

All models are implemented in PyTorch and trained using the Adam optimizer with learning rate scheduling and early stopping. Gradient clipping is applied for stability. Evaluation metrics include MAE, RMSE, and MAPE, computed over the prediction horizon.

3 Results

3.1 Quantitative Results

Table 2: Performance Comparison on METR-LA

Model	MAE	RMSE
MLP	5.23	12.59
NLSTM	5.09	12.57
Seq2Seq	5.08	12.69
DA-RNN	5.15	12.75
SeqGNN (Proposed)	5.08	12.11

SeqGNN achieves a good performance compared to other models due to effectiveness of graph-based modeling but the difference is less than expected.

3.2 Qualitative Analysis

SeqGNN captures congestion propagation patterns aligned with road connectivity. Unlike temporal-only models, it accurately predicts downstream congestion effects, particularly during peak traffic periods.

4 Discussion

The results confirm that incorporating graph topology significantly improves prediction accuracy but for this dataset the results does not differ by a long margin. SeqGNN outperforms all baselines by leveraging relational inductive biases inherent in road networks. While attention-based temporal models perform well, they lack explicit structural awareness.

Limitations include higher computational cost and reliance on accurate graph construction. Nevertheless, the performance gains justify the added complexity in large-scale urban applications.

5 Challenges, Limitations, and Future Work

- **Limitation:** Increased training time due to graph operations.(and GPU memory needed to trian the model effectively is very large)
- **Limitation:** Static graph assumption ignores dynamic road changes.
- **Future Work:** Integrate weather and event data.
- **Future Work:** Scale graphs with thousands of nodes.

6 Conclusion

This project demonstrates that Sequential Graph Neural Networks provide a powerful framework for traffic speed prediction but they can perform differently based on the nature of dataset. By modeling both temporal evolution and network topology, SeqGNN can outperforms traditional deep learning baselines. The results highlight the importance of graph mining techniques in spatiotemporal forecasting and validate SeqGNN as an effective approach for real-world intelligent transportation systems but other structure aware models like GAT or DCGRN should be experimented further to decide what model fits the best.

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